Unit 3 Guided Notes — Equations & Inequalities

ANSWER KEY

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

🎯 **By the end of this unit, I can…**

- Solve multi-step equations, including ones with variables on both sides
- Distribute and combine like terms accurately
- Identify equations with one solution, no solution, or infinitely many

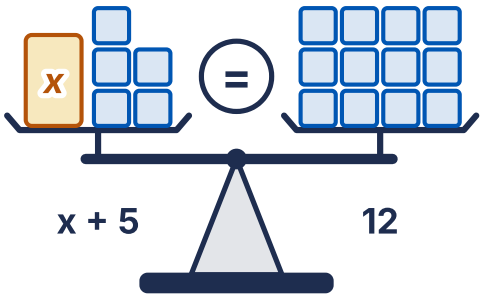
Standards: NC.8.EE.7

Answer key: every blank and final answer is filled in green.

1 One-Step Equations / Inverse Operations

📌 **THE BIG IDEA:** INVERSE OPERATIONS are used to isolate the variable. Whatever you do to one side, you MUST do to the other to keep the equation balanced!

x tile +1 tile red = negative (−x or −1) a big block = that number



x + 5 = 12 is balanced. Keep it balanced!

Operation	Inverse	Example
Addition	Subtraction	x + 5 = 12
Subtraction	Addition	x − 3 = 8
Multiplication	Division	3x = 15
Division	Multiplication	x/4 = 2

Match the Operation to its Inverse

MATCH Write the letter of each operation's inverse on the line.

B 1. Addition

D 2. Subtraction

A 3. Multiplication

C 4. Division

A. Division

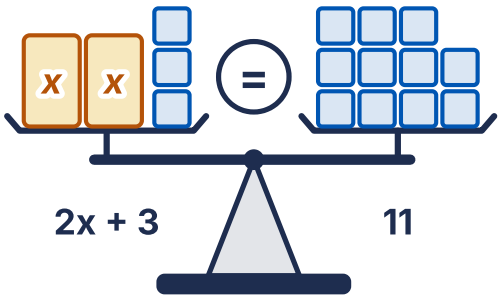
B. Subtraction

C. Multiplication

D. Addition

2 Two-Step Equations

📌 **TWO STEPS:** Step 1 — Undo addition/subtraction. Step 2 — Undo multiplication/division.



Whatever you do to one side, do the SAME to the other

Example 1: $-2x + 4 = 8$

$$-2x + 4 = 8$$

Step 1: Subtract 4 from both sides (undo the + 4)

Step 2: Divide both sides by −2

Answer: x = −2

Example 2: $x/2 + 6 = 7$

$$\frac{x}{2} + 6 = 7$$

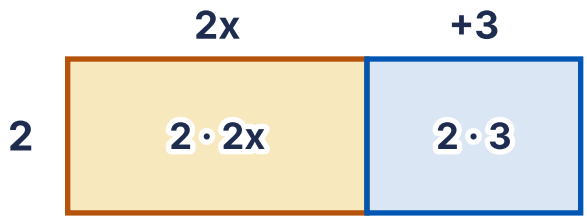
Step 1: Subtract 6 from both sides (undo the + 6)

Step 2: Multiply both sides by 2 (undo the ÷ 2)

Answer: x = 2

3 Distributive Property

📌 **THE RULE:** Any number (or variable) outside the parentheses is multiplied by EVERYTHING inside. Pay close attention to positive/negative signs!



The 2 multiplies BOTH parts

Example: $2(2x + 3) = -6$

$$2(2x + 3) = -6$$

Step 1: Distribute: multiply 2 by EACH term inside the ( )

Step 2: Subtract 6 from both sides (undo the + 6)

Step 3: Divide both sides by 4

Answer: $x = -3$

ERROR ANALYSIS A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?

A negative times a negative is positive: $-3 \cdot (-4) = +12$. Correct: $-3x + 12$. Distribute the sign to EVERY term inside the parentheses.

4 Variables on Both Sides

📌 **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: $5x + 3 = 2x + 12$

$$5x + 3 = 2x + 12$$

Step 1: Subtract 2x from both sides (gets the x's on one side)

Step 2: Subtract 3 from both sides (undo the + 3)

Step 3: Divide both sides by 3

Answer: x = 3

5 Multi-Step Equations — The 5-Step Process

📌 **THE 5 STEPS:** 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

$$-2x + 1 = 8(x + 2) - 5x$$

Step 1: Distribute: multiply 8 by EACH term inside the ( )

Step 2: Combine like terms: 8x − 5x = 3x

Step 3: Add 2x to both sides (gets the x's on one side)

Step 4: Subtract 16 from both sides (undo the + 16)

Step 5: Divide both sides by 5

Step 6: Read it with x first

Answer: x = −3

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like 5 = 9).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like 6 = 6).

Result	Meaning	Example
Variables stay; x = number	One Solution	2x+1=7 → x=3
Variables cancel; FALSE remains	No Solution	2x+1=2x+5 → 1=5 ×
Variables cancel; TRUE remains	Infinite Solutions	2x+1=2x+1 → 1=1 ✓

No Solution — Solve: 3(x + 2) = 3x + 10

$$3(x + 2) = 3x + 10$$

Step 1: Distribute: multiply 3 by EACH term inside the ( )

Step 2: Subtract 3x from both sides (gets the x's on one side)

Answer: No solution

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

$$2(x + 4) = 2x + 8$$

Step 1: Distribute: multiply 2 by EACH term inside the ( )

Step 2: Subtract 2x from both sides (gets the x's on one side)

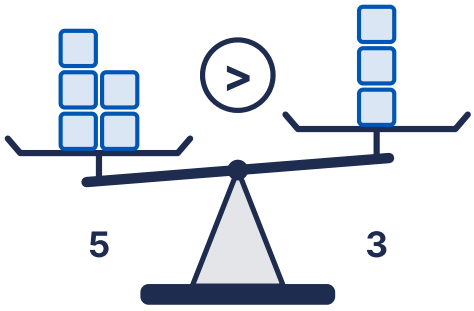
Answer: Infinitely many solutions

Quick Check

MC Solving an equation gives 7 = 7. The equation has...

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

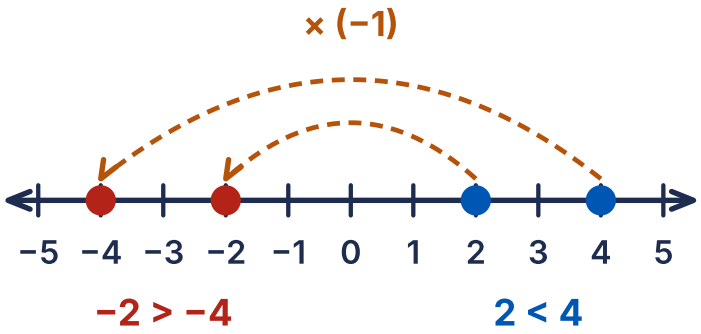
7 Linear Inequalities



An inequality is a scale that is NOT level

Symbol	Meaning	Circle	Shade
<	less than	OPEN	LEFT
>	greater than	OPEN	RIGHT
≤	less than or equal	CLOSED	LEFT
≥	greater than or equal	CLOSED	RIGHT

⚠️ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: −2x > 6 → divide by −2 → x < −3 (sign flipped!)



The order flips, so the sign flips.

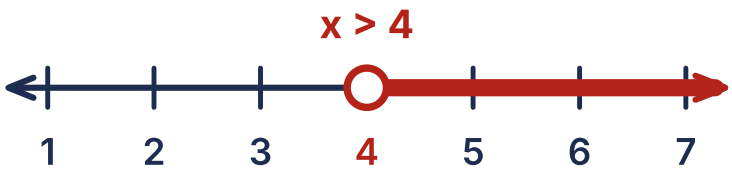
Example (no flip): 2x + 3 > 11

$$2x + 3 > 11$$

Step 1: Subtract 3 from both sides (undo the + 3)

Step 2: Divide both sides by 2

Answer: x > 4



x > 4: open circle (4 is not included)

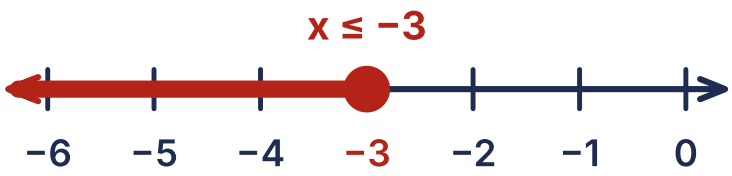
Example (FLIP NEEDED): $-3x + 4 \geq 13$

$$-3x + 4 \geq 13$$

Step 1: Subtract 4 from both sides (undo the + 4)

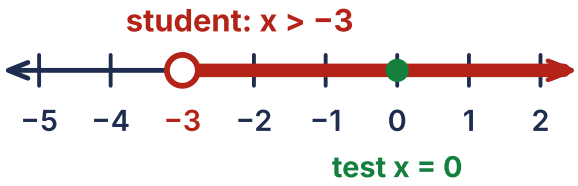
Step 2: Divide both sides by −3 — it is NEGATIVE, so FLIP the sign!

Answer: $x \leq -3$



x ≤ −3: closed circle (sign flipped!)

ERROR ANALYSIS To solve $-4x > 12$ a student wrote $x > -3$. What's the mistake?



Test $x = 0$ in $-4x > 12$. Is $-4(0) > 12$ true?

Dividing both sides by the negative number -4 requires FLIPPING the sign. Correct: $x < -3$.

8 Word Problems — Equations & Inequalities

📌 **THE FLOWCHART:** 1) READ → 2) DEFINE the variable → 3) IDENTIFY keywords → 4) WRITE the equation/inequality → 5) SOLVE → 6) CHECK → 7) ANSWER in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least (≥), at most (≤), more than (>), fewer than (<)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by



A bar model helps you see the equation: $6 fee + $4 for each ticket = $30 total.

SHORT ANSWER Movie Theater: A movie theater charges a $6 entry fee plus $4 per ticket. If the total cost was $30, how many tickets were purchased?

Let x = number of tickets

Equation: 6 + 4x = 30

Solve (show each step):

Answer: 6 tickets were purchased.

SHORT ANSWER Notebooks: A student wants to spend no more than $50 on notebooks that cost $6 each. How many notebooks can the student buy?

Let x = number of notebooks

Equation: 6x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so the student can buy **at most 8 notebooks.**

✏ Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Multi-Step Equations

1) $3x + 7 = 22$

Answer: $x = 5$

2) $2(x - 4) = 10$

Answer: $x = 9$

3) $5x - 3 = 2x + 9$

Answer: $x = 4$

4) $-4(x + 2) = 12$

Answer: $x = -5$

5) $6x + 4 = 4x - 8$

Answer: $x = -6$

6) $3(2x + 1) = 4x + 7$

Answer: $x = 2$

7) $x/4 + 5 = 8$

Answer: $x = 12$

8) $7 - 2x = 3x + 17$

Answer: $x = -2$

9) 0.5x + 3 = 7

Answer: x = 8

10) −(x − 6) = 2x + 3

Answer: x = 1

B. Special Cases — No Solution or Infinite Solutions?

1) 2x + 3 = 2x + 3

Answer:
Infinitely many solutions

2) 4x + 1 = 4x − 5

Answer:
No solution

3) 3(x + 2) = 3x + 6

Answer:
Infinitely many solutions

4) 5x − 5 = 5(x + 1)

Answer:
No solution

5) 7x + 2 = 7x + 2

Answer: Infinitely many solutions

6) x + 4 = x − 1

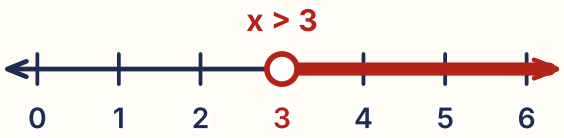
Answer: No solution

C. Inequalities — Solve

Solve it like an equation. FLIP the sign only when you multiply or divide both sides by a negative number.

1) 3x + 2 > 11

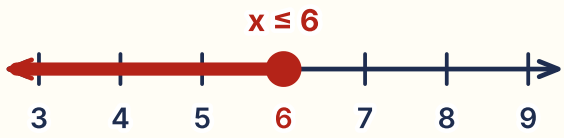
Answer: x > 3



x > 3: open circle at 3, shade right

2) 2x − 5 ≤ 7

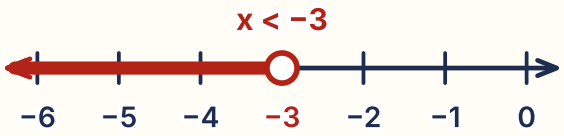
Answer: x ≤ 6



x ≤ 6: closed circle at 6, shade left

3) $-4x > 12$ (flip!)

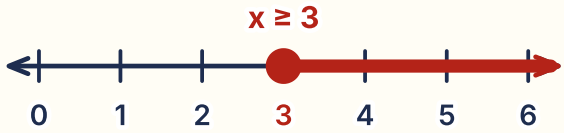
Answer: $x < -3$



$x < -3$: open circle at -3, shade left

4) $5x + 1 \geq 16$

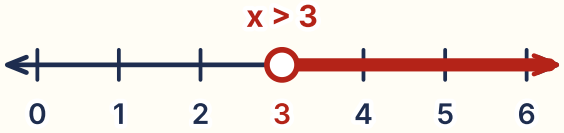
Answer: $x \geq 3$



$x \geq 3$: closed circle at 3, shade right

5) $7 - x < 4$ (flip!)

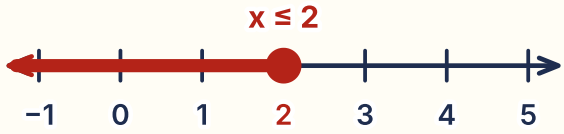
Answer: $x > 3$



$x > 3$: open circle at 3, shade right

6) $2(x + 3) \leq 10$

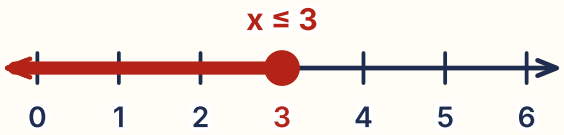
Answer: $x \leq 2$



$x \leq 2$: closed circle at 2, shade left

7) $-3x + 9 \geq 0$ (flip!)

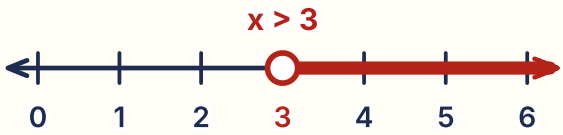
Answer: $x \leq 3$



x ≤ 3: closed circle at 3, shade left

8) $4x - 1 > 2x + 5$

Answer: $x > 3$



x > 3: open circle at 3, shade right

D. Word Problems

1. A phone plan costs $35/month plus $0.10 per text. Last month's bill was $52. How many texts were sent?

Let x = number of texts

Equation: $35 + 0.10x = 52$

Solve (show each step):

Answer: 170 texts were sent.

2. Sam has $240 and saves $30/week. Jake has $420 and spends $10/week. When will they have the same amount?

Let x = number of weeks

Equation: 240 + 30x = 420 − 10x

Solve (show each step):

Answer: They will have the same amount after 4.5 weeks.

3. A rectangle's perimeter is 56 cm. The length is 4 cm more than the width. Find the width.

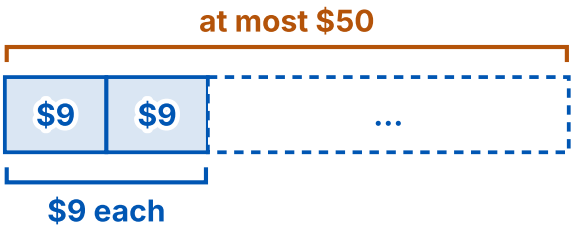
Let x = width (cm), so the length is x + 4

Equation: 2(x + 4) + 2x = 56

Solve (show each step):

Answer: The width is 12 cm.

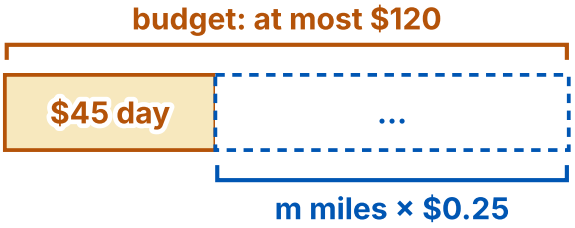
4. Movie tickets cost $9 each. You have at most $50 to spend. What is the maximum number of tickets?



How many $9 tickets fit in $50?

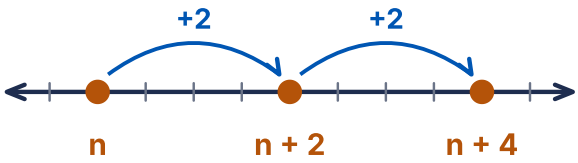
Let t = tickets → 9t ≤ 50 ("at most" = ≤). Solve: t ≤ 5.56 → max 5 tickets (must be a whole number).

5. A car rental costs $45/day plus $0.25/mile. Your budget is $120 for one day. How many miles can you drive?



Let m = miles → 45 + 0.25m ≤ 120. Solve: 0.25m ≤ 75 → m ≤ 300 miles.

6. Three consecutive even integers sum to 78. Find the smallest integer.



n + (n + 2) + (n + 4) = 78

Let n = smallest → n + (n+2) + (n+4) = 78. Solve: 3n + 6 = 78 → n = 24.

← Back to Unit 3